

Github Action

GitHub Actions is currently one of the most popular CI/CD tools. You can automate build and deployment processes simply by configuring a few files within your GitHub repository, without the need to set up complex servers.

GitGithub.pdf	Introduce Git CLI(Command Line Interface) and Github.
CppUTest.pdf	Introduce Unit Test with CppUTest and CppUMock. You can download a sample routine at " https://github.com/vspauto/utest.git "
Docker.md	Explain about Docker

Prepare

```
$ cd ~/github
$ git clone https://github.com/vspauto/utest.git
$ cd utest
```

.gitignore

```
# C/C++ output
*.o
*.obj
*.out
*.exe

# CppUTest framework
cpputest/
*.tar.gz
```

Docker

~/github/utest/Dockerfile

```
# Fetch ubuntu image
FROM ubuntu:22.04

# Install build tools
RUN apt update && \
    apt install -y wget build-essential autoconf automake libtool

# Copy project into image
RUN mkdir /project
COPY src /project/src
```

```

COPY tests /project/tests
COPY Makefile /project/Makefile

# Download and build CppUTest
RUN mkdir /project/tools/ && \
  cd /project/ && \
  wget https://github.com/cpptest/cpptest/releases/download/v4.0/cpptest-4.0.tar.gz && \
  tar xf cpptest-4.0.tar.gz && \
  mv cpptest-4.0/ cpptest/ && \
  cd cpptest/ && \
  autoreconf -i && \
  ./configure && \
  make

# Execute script
ENTRYPOINT ["make", "test", "-C", "/project/"]

```

Open “docker desktop”, and then execute below on Terminal.

```

$ cd ~/github/utest
$ docker build -t utest-image -f Dockerfile .

```

```

$ docker run --rm utest-image

```

```

Running average_tests
..
OK (2 tests, 2 ran, 2 checks, 0 ignored, 0 filtered out, 0 ms)

Running mock_sensor_tests
..
OK (2 tests, 2 ran, 7 checks, 0 ignored, 0 filtered out, 0 ms)

Running ring_buffer_tests
...
OK (3 tests, 3 ran, 15 checks, 0 ignored, 0 filtered out, 0 ms)

```

Github Action

Prepare your YAML file on “.github/workflows” directory

```

~/github/utest/
+-- .github/
    +-- workflows/
        +-- utest.yml

```

```

name: Unit Tests

on: [push]

jobs:
  docker-unit-tests:
    runs-on: ubuntu-latest
    name: Build and run unit tests Docker image
    steps:
      - name: Check out this repo
        uses: actions/checkout@v3

      - name: Build Docker image
        run: docker build -t utest-image -f Dockerfile .

      - name: Run image
        run: docker run --rm utest-image

```

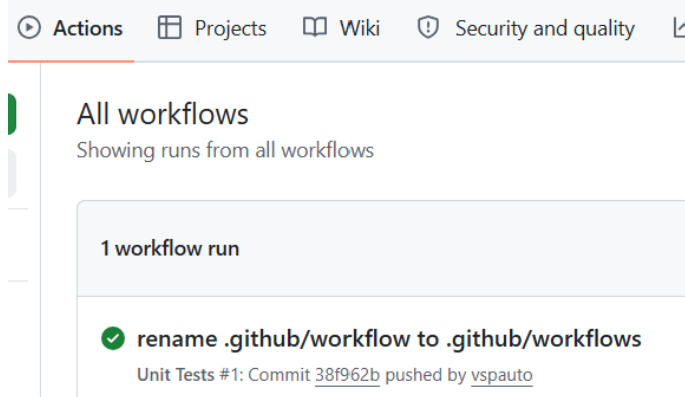
Push to Github.

```

$ git add .github/
$ git commit -m "add .github/workflow/utest.yml"
$ git push -u origin main

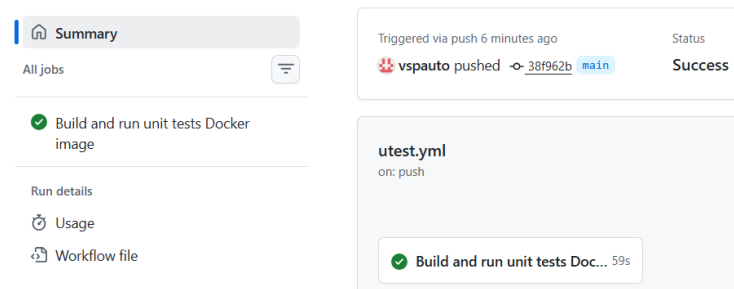
```

On your Github repository, select "Action". It will build your workflow.



Select your workflow.

✓ rename .github/workflow to .github/workflows #1



Select "Build and run unit tests Docker image"

← Unit Tests

✓ rename .github/workflow to .github/workflows #1

The screenshot shows the GitHub Actions interface for a workflow. On the left, there's a sidebar with 'Summary' and 'All jobs'. The 'All jobs' section shows a single job 'Build and run unit tests Docker image' with a green checkmark. Below this, there are links for 'Run details', 'Usage', and 'Workflow file'. The main area shows the job details for 'Build and run unit tests Docker image', which succeeded 5 minutes ago in 59s. Below the job title, there's a list of steps: 'Set up job', 'Check out this repo', 'Build Docker image', 'Run image' (highlighted), 'Post Check out this repo', and 'Complete job'. Each step has a green checkmark.

Select "Run image". It will show below test results.

```
..  
Running average_tests  
..  
OK (2 tests, 2 ran, 2 checks, 0 ignored, 0 filtered out, 0 ms)  
  
..  
Running mock_sensor_tests  
..  
OK (2 tests, 2 ran, 7 checks, 0 ignored, 0 filtered out, 1 ms)  
..  
Running ring_buffer_tests  
...  
OK (3 tests, 3 ran, 15 checks, 0 ignored, 0 filtered out, 0 ms)
```

reference:

<https://github.com/ShawnHymel/c-unit-test/tree/main>

<https://www.youtube.com/watch?v=IZWFmEhlhpY&t=2708s>